FOUNDATIONS OF ARTIFICIAL INTELLIGENCE

PROJECT REPORT

AI BASED BANK LOAN PRE-SCREENING AND DESCISION MAKING SYSTEM

A PROJECT REPORT

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ABSTRACT

Evaluating bank loan applications requires assessing applicants' financial details, employment status, credit history and repayment capacity against relevant eligibility conditions. Manual initial screening can be time-consuming and may make it difficult to provide consistent explanations for decisions. This project presents the AI-Based Bank Loan Pre-Screening and Decision-Making System, named LoanSense AI, which applies a Rule-Based Expert System to automate the initial evaluation of loan applications. The system uses IF–THEN rules and a rule-based inference engine to assess applicant information for four loan categories: Personal Loan, Home Loan, Car Loan and Education Loan. It validates the submitted details, identifies missing or invalid information, and evaluates the configured eligibility conditions to generate one of three outcomes: Eligible, Ineligible or Incomplete. Each outcome includes a readable explanation of the evaluated conditions, the reasons for the result and any additional information required. The system also provides user registration, login and application history, while an SQLite database stores application details and pre-screening results for future reference. The application is developed using HTML5, CSS3, JavaScript, Bootstrap 5, Python Flask and SQLite. The project demonstrates the use of Artificial Intelligence concepts, including knowledge representation, rule-based reasoning, inference and explainable decision-making, in the initial screening of loan applications. It is designed as an educational prototype and does not represent official SBI loan approval or sanction.

Keywords: Artificial Intelligence, Rule-Based Expert System, Loan Pre-Screening, IF–THEN Rules, Inference Engine, Eligibility Evaluation, Python Flask, SQLite.

CHAPTER 1

INTRODUCTION

1.1 Background

Banks receive many loan applications from individuals with different income levels, employment statuses and financial backgrounds. Before processing a loan further, the applicant's information must be checked against the eligibility conditions of the selected loan scheme. Manual initial screening can be time-consuming, and identifying missing details or explaining decisions may become difficult.

When this process is performed manually, checking eligibility conditions and identifying incomplete information can require considerable effort. Different loan categories also have different requirements. Personal loans may depend on salary and employment conditions, home loans on income and repayment capacity, car loans on applicant category and income, and education loans on admission status and course details.

Loan pre-screening is a suitable application of Artificial Intelligence. A Rule-Based Expert System uses a knowledge base containing IF–THEN rules and an inference engine to evaluate information and produce a decision. This project applies these concepts to develop the AI-Based SBI Multi-Loan Eligibility Checker, an educational application that evaluates applicant details using configured loan-specific rules and provides understandable pre-screening results.

1.2 Problem Statement

Evaluating loan applications manually can be time-consuming and may lead to inconsistent assessments. Different loan categories have different eligibility conditions that must be evaluated carefully. The proposed system uses IF–THEN rules to assess applicant details and validate the submitted information. It identifies missing details and generates Eligible, Ineligible or Incomplete results. The system provides clear explanations for each result and stores application records in SQLite for future reference.

1.3 Objectives

- To develop an AI-based loan pre-screening system using a Rule-Based Expert System and IF–THEN rules.
- To evaluate eligibility conditions for Personal, Home, Car and Education Loans.
- To validate applicant details and identify missing or invalid information.

- To explain the reasons behind each eligibility result.
- To provide secure user registration, login and session management.
- To store application details and results in SQLite for future reference.
- To provide a simple interface with loan selection, eligibility evaluation and application history.

1.4 Scope and Contributions

The system covers four major SBI loan categories: Personal Loan, Home Loan, Car Loan and Education Loan. It evaluates applicant information using configured eligibility conditions and provides understandable pre-screening results. The main contributions of this work are:

- A rule-based inference engine that evaluates loan eligibility using IF–THEN rules and a structured knowledge base.
- A loan-specific rule repository containing eligibility conditions, official SBI source links and verification statuses.
- An explanation mechanism that identifies passed, failed and unresolved rules and provides clear reasons for each result.
- A secure web application with user authentication, SQLite-based application storage and user-specific history.

The rest of this report is organised as follows. Chapter 2 reviews related work on loan eligibility assessment and rule-based expert systems. Chapter 3 describes the proposed system, its architecture, database design and inference algorithm. Chapter 4 explains the individual modules and their functions. Chapter 5 presents the implementation environment, test cases, results and application screenshots. Chapter 6 concludes the report and discusses possible future enhancements.

CHAPTER 2

LITERATURE REVIEW

This chapter reviews existing work on loan eligibility assessment, rule-based expert systems, automated decision-making and explainable Artificial Intelligence. These areas provide the foundation for developing the AI-Based Multi-Loan Eligibility Checker.

2.1 Loan Eligibilty Assessment

Loan eligibility assessment involves evaluating an applicant's financial details, income, employment status and repayment capacity against specific loan conditions. Manual screening can be time-consuming because different loan categories have different eligibility requirements. Personal loans may depend on salary and employment conditions, while home and car loans consider income and repayment obligations. Education loans involve course details, admission status and institution category. A structured system helps evaluate applications using the relevant loan-specific conditions and provides consistent pre-screening results.

2.2 Rule-Based expert systems

Rule-based expert systems are an established Artificial Intelligence approach for representing knowledge and solving problems using predefined rules. Buchanan and Shortliffe (1984) described the development of knowledge-based expert systems and demonstrated how domain knowledge can be represented to support decision-making.

Russell and Norvig (2020) explain knowledge representation and logical reasoning as important areas of Artificial Intelligence. In a rule-based system, knowledge can be expressed through IF–THEN statements. For example, if a mandatory eligibility condition is satisfied, the corresponding rule passes; if a verified mandatory condition is not satisfied, the rule fails.

The proposed project uses this approach to represent loan-specific eligibility conditions in a knowledge base. An inference engine evaluates the applicable rules against the applicant's submitted information and produces a result based on the evaluation outcomes.

2.3 Automated Decision Making and Interfaces

Automated decision-making systems apply predefined conditions to input data to produce consistent results. In expert systems, the inference engine examines the available facts and determines which rules are satisfied, violated or remain unresolved. Giarratano and Riley (2005) discuss the architecture and reasoning mechanisms of expert systems, including knowledge bases and inference engines.

In loan pre-screening, automated rule evaluation can help identify missing information and explain why an application satisfies or fails a configured condition. However, a rule-based system must distinguish between verified requirements and conditions that have not been confirmed. This project uses three rule outcomes: Pass, Fail and Not Evaluated. The final result is classified as Eligible, Ineligible or Incomplete according to the applicable rules and available information.

2.4 Explainable AI in Loan Screening

Explainability is important in decision-support applications because users need to understand how a result was generated. A system that displays only a final decision may not help applicants identify missing details or understand which conditions affected the outcome.

Guidotti et al. (2018) reviewed methods for explaining decisions produced by complex predictive systems and highlighted the importance of understanding model behaviour. Although the proposed project uses explicit rules rather than a machine-learning model, the same need for understandable decisions applies. Its explanation module presents the rules passed, mandatory conditions failed and information requiring further verification.

The system also uses official SBI scheme pages as references when configuring eligibility rules. Conditions that cannot be verified are labelled accordingly instead of being treated as confirmed bank policy. This improves transparency and prevents the educational prototype from presenting its results as official loan approval or rejection.

2.5 Summary and Research Gap

Table 2.1 Comparison of approaches

Approach	Strength	Limitation	Use in this project
Manual Screening	Detailed Assessment	Time-consuming	Basis for automated Screening
Online loan form assessment	Easy data collection	Limited decision support	Collects applicant details
Rule Based Expert System	Consistent and explainable results	Depends on defined rules	Evaluates loan eligibility
Proposed Multi-Loan Eligibility Checker	Loan-specific results and explanations	Uses configured eligibility conditions	Screens Personal, Home, Car and Education Loans

The proposed system addresses this gap by integrating a knowledge base, rule-based inference engine, explanation generator and SQLite database into one web application. It supports four loan categories and provides understandable results while identifying missing information and unverified eligibility conditions. The project demonstrates the practical application of Artificial Intelligence reasoning without relying on machine learning or claiming to make official SBI lending decisions .

CHAPTER 3

PROPOSED SYSTEM

The proposed AI-Based Bank-Loan Eligibility Checker is a full-stack web application designed to perform educational loan pre-screening using a Rule-Based Expert System. The frontend is developed using HTML5, CSS3, JavaScript and Bootstrap 5. The backend uses Python with Flask, while SQLite stores user details and application records

3.1 Architecture Diagram

Figure 3.1 shows the overall architecture of the AI-Based Bank Loan Eligibility Checker. Users access the web application to register, log in, select a loan category and enter their financial details. The system evaluates the submitted information using a rule-based inference engine and displays the pre-screening result with explanations.

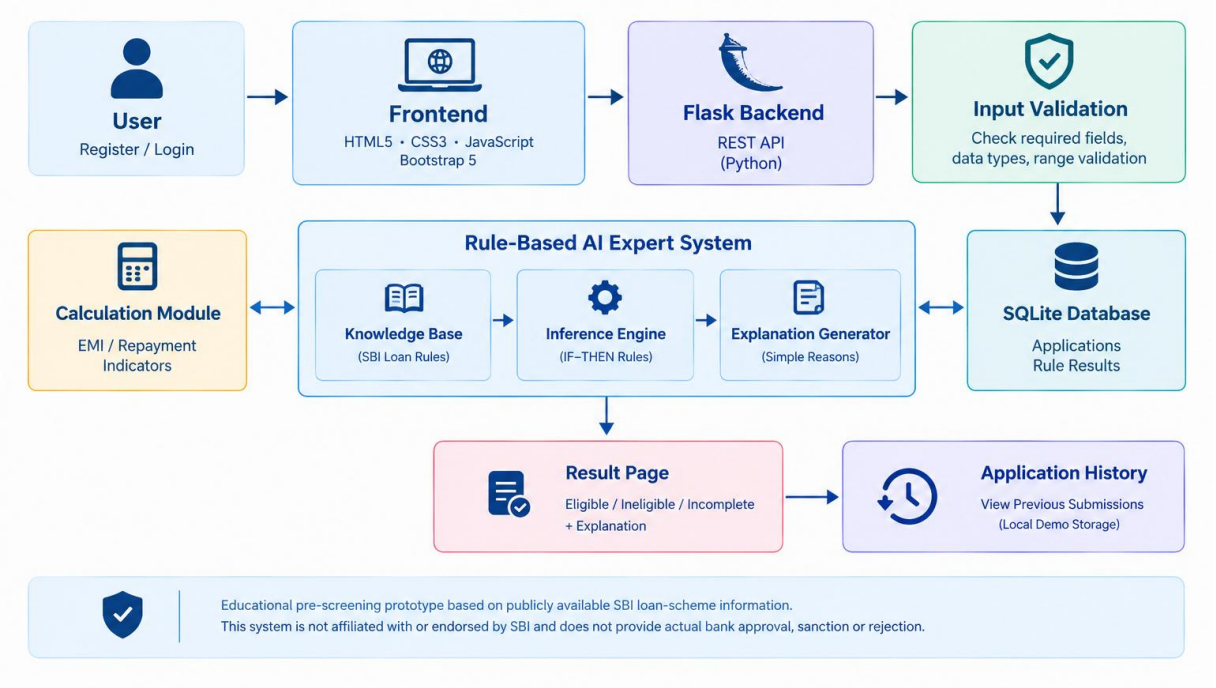


Fig. 3.1 System architecture of the AI based bank loan eligibility checker

The architecture has four layers:

- **Presentation Layer**: HTML5, CSS3, JavaScript and Bootstrap 5 provide the user interface for registration, login, loan selection, application forms and result pages.

- **Application Layer:** Python Flask handles user authentication, session management, input validation and communication between the frontend, inference engine and database.
- **Intelligence layer:** The rule-based inference engine evaluates loan eligibility using IF–THEN rules. The knowledge base stores loan conditions, while the explanation module provides reasons for the generated results.
- **Data layer:** SQLite stores user account details, loan application information and pre-screening results, allowing users to access their application history.

Table 3.1 Technology stack

Layer	Technology	Purpose
Frontend	HTML5, CSS3, JavaScript, Bootstrap 5	User interface and responsive design
Backend	Python, Flask	Application logic and request handling
Security	Password hashing, Flask sessions	Secure login and session management
Database	SQLite	Store user and application records
AI Engine	Rule-Based Expert System	Evaluate loan-specific IF–THEN rules

3.2 Database Design

The SQLite database stores user details, loan application information and pre-screening results. It supports secure user access and allows applicants to view their previous applications. The database is organised into related tables to maintain records efficiently.

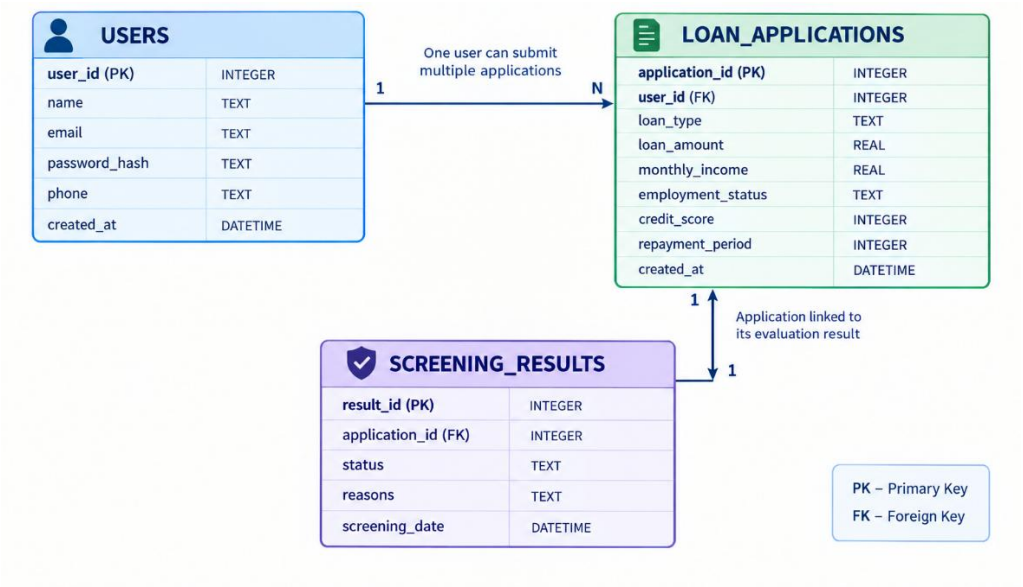


Fig. 3.2 Logical data model (key fields of each collection)

The main database tables are:

- **Users**: Stores user ID, name, email, hashed password and account details.
- **Loan Applications**: Stores the applicant ID, loan category, requested amount, income, employment status, credit score and repayment period.
- **Screening Results**: Stores the application ID, result status, evaluation reasons and screening date.

3.3 Rule Based Inference Formulation

The proposed system uses a Rule-Based Expert System to evaluate loan applications. The knowledge base contains configured eligibility conditions for each supported loan category. The inference engine compares the applicant's details against the relevant rules and generates a pre-screening result.

Knowledge base

The knowledge base stores IF–THEN rules related to income, employment status, credit score, requested loan amount, repayment period and repayment capacity. The applicable conditions depend on the selected loan category. The rules used in this educational prototype must be documented and verified against available SBI scheme information before being described as official requirements.

Table 3.2 Supported Loan Categories

S. No.	Loan Category	Details Evaluated
1	Personal Loan	Income, employment and repayment capacity
2	Home Loan	Income, loan amount and repayment obligations
3	Car Loan	Income, vehicle loan amount and repayment capacity
4	Education Loan	Course details, admission information and repayment-related details

Table 3.3 Rule Evaluation Outcomes

Outcome	Condition	System Response
Eligible	All applicable configured rules are satisfied	Displays a positive pre-screening result
Ineligible	One or more defined disqualifying rules are triggered	Displays the failed conditions and reasons
Incomplete	Required details are missing or invalid	Requests the missing or corrected information

These outcomes represent the system's educational pre-screening assessment and do not indicate an actual SBI sanction or rejection

3.4 Rule Based Inference Process

The inference engine processes the applicant's details through a sequence of validation and rule-evaluation steps. Figure 3.3 illustrates the process.

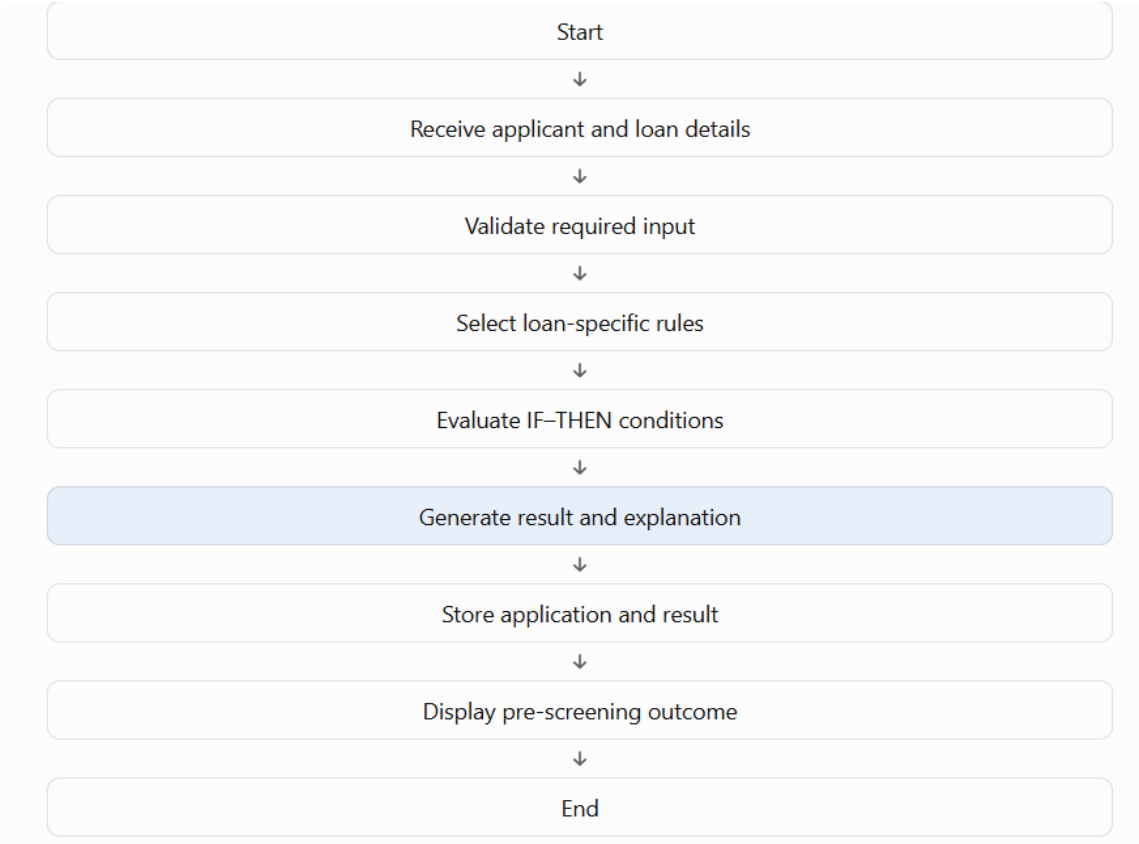


Fig. 3.3 Rule Based Inference Process

Algorithm 3.1: Rule-Based Loan Pre-Screening

1. Start the system and receive the applicant's details.
2. Validate the input fields and identify missing or invalid information.
3. If required details are incomplete, request corrections and return an Incomplete result.
4. Select the knowledge base corresponding to the chosen loan category.
5. Apply the relevant IF–THEN rules to evaluate the applicant's details.
6. Determine whether the configured conditions are satisfied or a disqualifying condition is triggered.
7. Generate the result with explanations of passed, failed or unresolved rules.
8. Save the application and screening result in SQLite.

9. Display the pre-screening result and end the process.

Rule-Based Evaluation Examples

The following examples illustrate the rule format used in the educational system. They are not official SBI eligibility thresholds.

- **Rule 1**: IF required applicant details are missing, THEN the result is Incomplete.
- **Rule 2**: IF a defined disqualifying condition is triggered, THEN the result is Ineligible under the configured rules.
- **Rule 3**: IF all applicable configured eligibility rules are satisfied, THEN the result is Eligible under the configured rules.
- **Rule 4**: IF an applicable condition cannot be evaluated, THEN mark that condition as unresolved and request the required information.

Why Rule-Based Inference Is Suitable

Rule-based inference is suitable for this project because loan eligibility conditions can be represented as understandable IF–THEN statements. The system evaluates these rules systematically, making the decision process easier to test and explain. Unlike a machine-learning model, it does not require a training dataset. However, its results depend on the correctness and completeness of the configured rules.

3.5 Verification and Explainability

The system verifies applicant inputs before performing the eligibility evaluation. It checks required fields, numeric values and loan-specific information to prevent incomplete or invalid data from being processed as a final result.

The explanation module records the outcome of each applicable rule and generates a human-readable explanation. For example, the result page may identify conditions that were satisfied, conditions that failed and details that remain unresolved. This helps applicants understand how the system reached its pre-screening outcome.

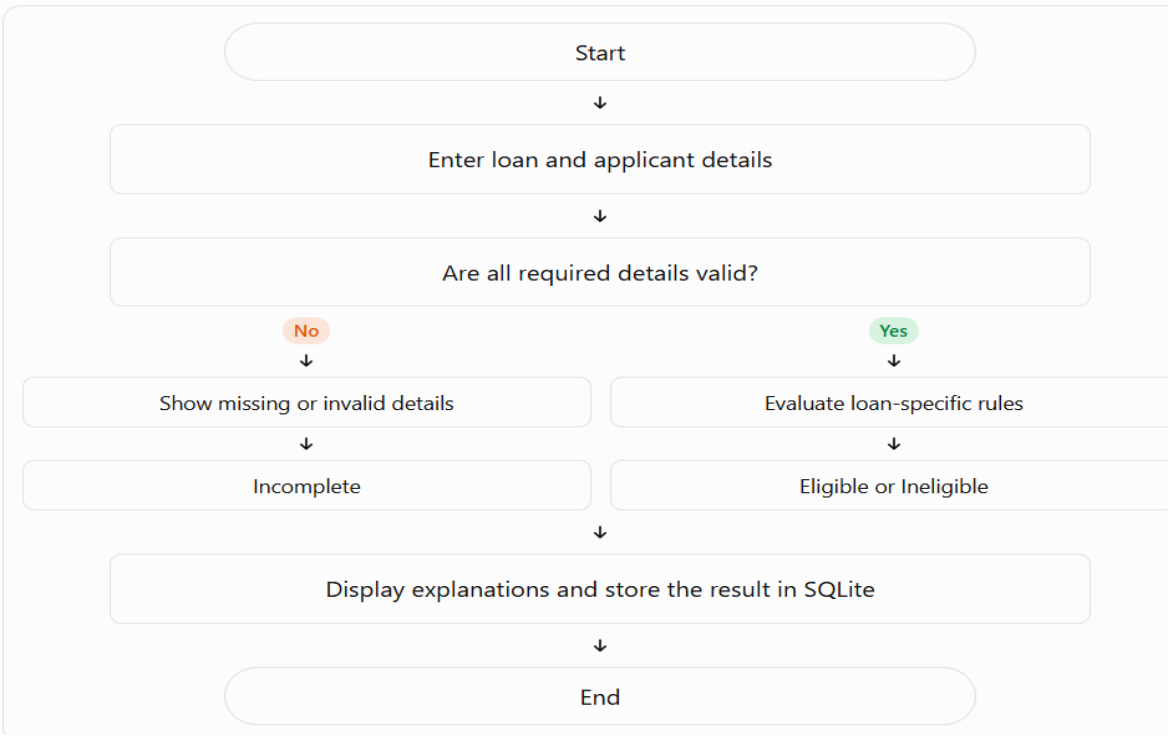


Fig 3.4 Loan Decision flowchart

The system also maintains application history so users can review previous screening results. Before a loan rule is presented as an SBI requirement, its source and verification status should be documented. This improves transparency and prevents the educational prototype from presenting unverified conditions as official bank policy. The final result is an explainable pre-screening estimate, not a guarantee of loan approval.

CHAPTER 4

MODULES DESCRIPTION

The AI-Based Bank Loan Eligibility Checker is divided into eight modules that work together to evaluate loan applications. The modules handle user authentication, loan selection, application entry, input validation, rule-based inference, explanation generation, result display and application history. Figure 4.1 presents the overall module structure of the proposed system.

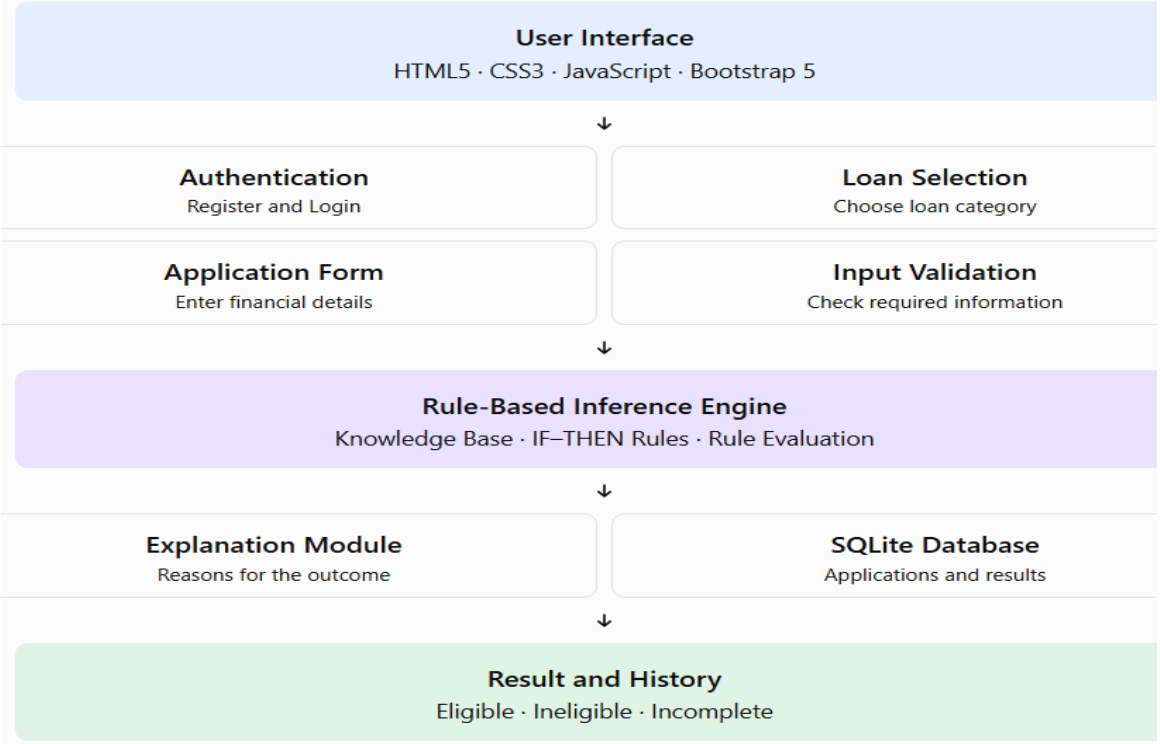


Fig. 4.1 Module overview of the system

4.1 Module 1: User Authentication

This module allows users to register and log in to the application. It verifies login credentials and manages user sessions to protect access to application pages. Passwords are stored using secure hashing rather than plain text.

- **Input**: Name, email and password.
- **Processing**: Validate registration details, hash passwords and verify login credentials.
- **Output**: Successful login and access to the appropriate pages.
- **Purpose:** Protects user accounts and supports individual application history.

4.2 Module 2: Loan Selection

This module allows users to choose the loan category they want to evaluate. Each category uses its corresponding form fields and configured eligibility rules.

- **Input**: Selected loan category.
- **Processing**: Identify the applicable loan type and load its required fields and rules.
- **Output**: A loan-specific application form.
- **Supported categories:** Personal Loan, Home Loan, Car Loan and Education Loan.

4.3 Module 3: Loan Application and Input Validation

This module collects the applicant's financial and loan-related details. The validation component checks required fields, data types, acceptable values and missing information before the inference engine runs.

- **Input:** Monthly income, employment status, credit score, requested loan amount and repayment period, along with other loan-specific details.
- **Processing**: Validate input fields and identify missing or invalid values.
- **Output**: Validated application data or a list of corrections required.
- **Purpose:** Reduces incorrect input and prevents incomplete information from being treated as a completed assessment.

Table 4.1 System Modules and their functions

S. No.	Module	Function
1	User Authentication	Handles registration, login and sessions
2	Loan Selection	Identifies the selected loan category
3	Application and Validation	Collects and validates applicant details
4	Knowledge Base	Stores configured loan eligibility rules
5	Inference Engine	Evaluates applicable IF–THEN rules
6	Explanation Module	Explains passed, failed and unresolved rules
7	Result Generation	Displays the pre-screening outcome
8	Database and History	Stores applications and retrieves past results

4.4 Module 4: Knowledge Base Management

The knowledge base stores the IF–THEN rules used to evaluate loan applications. Each loan category has its own set of configured eligibility conditions based on relevant applicant and loan information. The rules must be checked against reliable sources before being represented as official bank requirements.

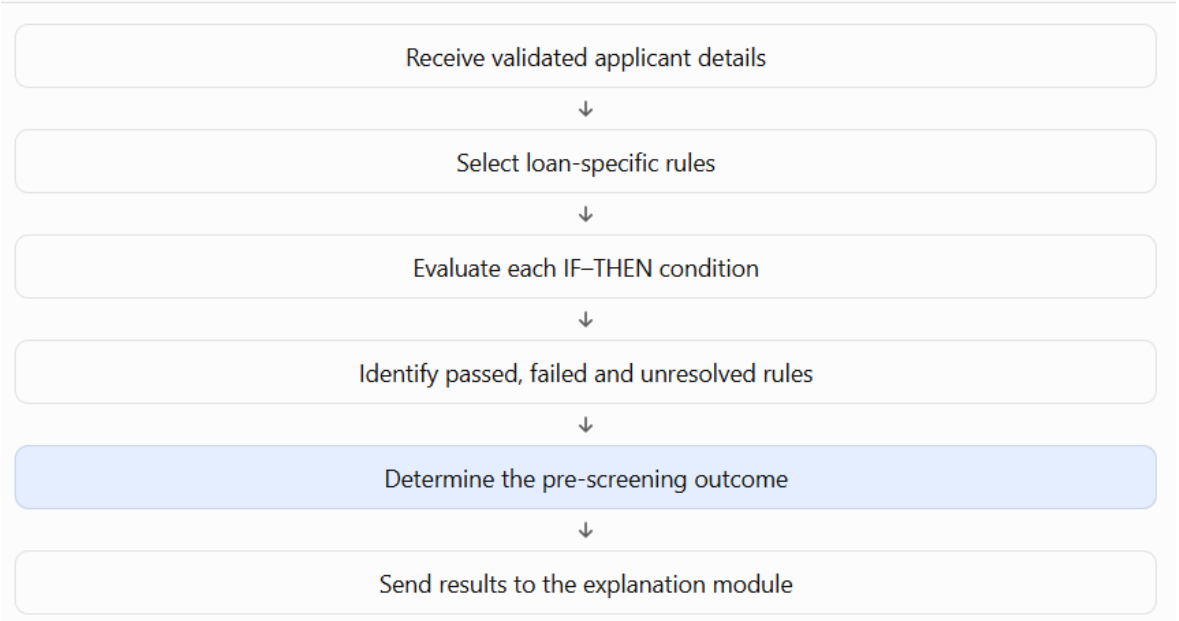
- **Input:** Loan category and configured eligibility conditions.
- **Processing:** Select the appropriate rules for the chosen loan type.
- **Output:** A set of applicable rules for evaluation.
- **Purpose:** Maintains structured and reusable eligibility knowledge.

4.5 Module 5: Rule-Based Inference Engine

The inference engine is the core AI component of the system. It compares validated applicant information with the applicable rules in the knowledge base. The engine identifies satisfied conditions, failed conditions and conditions that cannot yet be evaluated.

- **Input:** Validated applicant details and applicable rules.
- **Processing**: Apply IF–THEN rules and evaluate each condition.
- **Output:** Rule evaluation results for the selected loan.
- **Purpose:** Produces a consistent pre-screening assessment without requiring a machine-learning training dataset.

Fig 4.2 rule Based Interference and Evaluation flow



4.6 Module 6: Explanation Generation

This module converts the inference engine's rule evaluation results into understandable explanations. It helps users identify which configured conditions were satisfied, which were not satisfied and which require additional information.

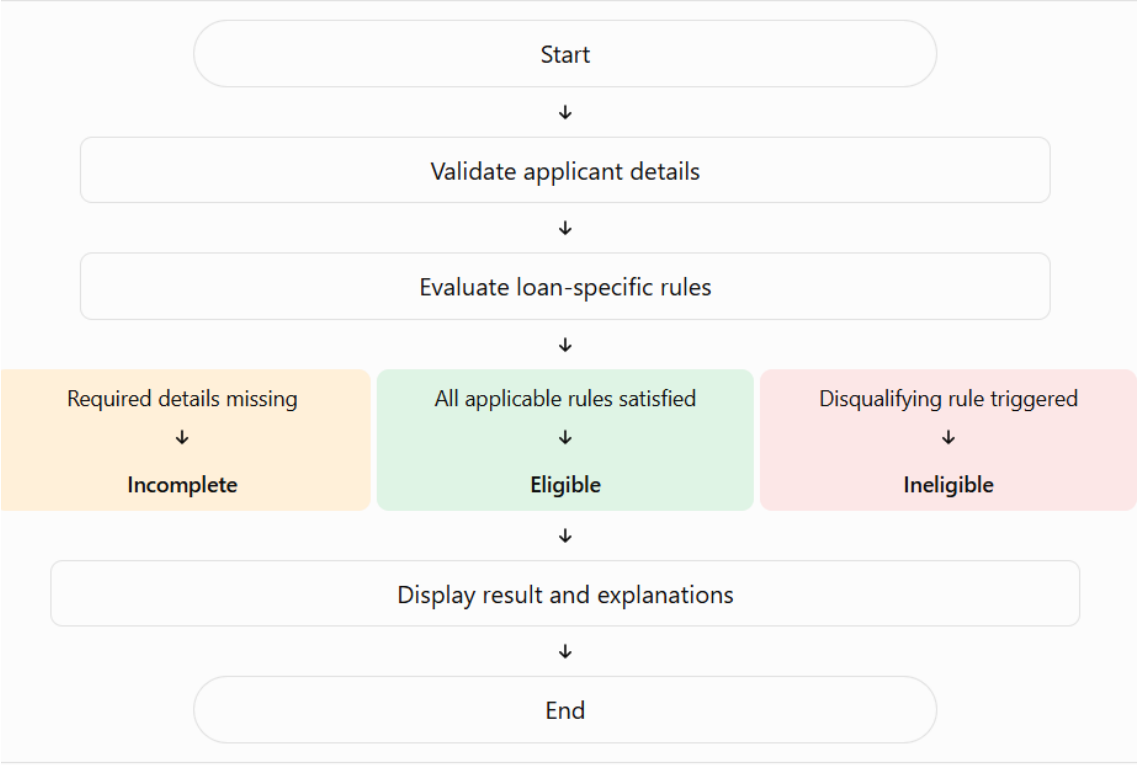
- **Input**: Rule evaluation results.
- **Processing:** Organise rule outcomes and generate explanations.
- **Output**: A clear explanation of the pre-screening result.
- **Purpose:** Improves transparency and helps users understand how the system reached its result.

4.7 Module 7: Pre-Screening Result Generation

This module displays the final result produced by the inference engine. The result can be Eligible, Ineligible or Incomplete according to the configured rules and the availability of required information. The result page also presents the relevant explanations.

- **Input:** Rule outcomes and validation status.
- **Processing**: Determine the result status and prepare the result summary.
- **Output:** Pre-screening status, reasons and unresolved conditions.
- **Purpose:** Provides an understandable result for the applicant.

Fig 4.3 Loan pre-screening decision flowchart



4.8 Module 8: Database and Application History

This module uses SQLite to store user information, submitted loan applications and screening results. It retrieves previous records for the logged-in user and displays the relevant application details and outcomes.

- **Input**: User details, application information and screening results.
- **Processing:** Save records, associate applications with users and retrieve history.
- **Output:** Stored application records and a history page.
- **Purpose:** Supports record keeping, review of previous assessments and continuity between user sessions.

Application History Workflow

1. The user submits a loan application through the web interface.
2. The inference engine evaluates the application using configured IF–THEN rules.
3. The system generates the pre-screening result with explanations.
4. The application details and screening result are stored in the SQLite database.
5. The user opens the Application History page after logging in.
6. The system retrieves the user's previous applications from the database.
7. The application history page displays the saved details and pre-screening results.

CHAPTER 5

IMPLEMENTATION AND RESULTS

5.1 Experimental Setup

The AI-Based Bank Loan Eligibility Checker was developed as a web application using HTML5, CSS3, JavaScript and Bootstrap 5 for the frontend, Python Flask for the backend and SQLite for data storage. The rule-based inference engine evaluates applicant details using configured IF–THEN rules.

Table 5.1 Software environment

Component	Version / tool
Operating System	Windows 10/11
Frontend	HTML5, CSS3, JavaScript, Bootstrap 5
Backend	Python, Flask
Database	SQLite
AI Technique	Rule-Based Expert System
Code Editor	Visual Studio Code

Table 5.2 Sample Applicant Test Dataset

The system uses sample applicant details to test the behaviour of the rule-based inference engine. The following table illustrates possible test inputs; replace them with the actual test cases used in your application.

Table 5.2 Sample application dataset

S. No.	Loan Type	Monthly Income	Employment Status	Test Purpose
1	Personal Loan	₹45,000	Salaried	Test normal rule evaluation
2	Home Loan	₹65,000	Salaried	Test home loan conditions
3	Car Loan	₹40,000	Self-employed	Test employment-related rules
4	Education Loan	₹20,000	Student / Applicant	Test education loan conditions
5	Personal Loan	Not provided	Salaried	Test missing information

5.2 Results

The proposed system provides a complete workflow for user authentication, loan selection, applicant detail submission, rule-based evaluation and result display. The following five screenshots can be included in this section to demonstrate the implemented application.

5.2.1 Login Page and registration page

Figure 5.1 shows the login and registration interface of the system. Users can create an account and log in to access the loan eligibility checker. The authentication module validates credentials and manages user sessions.

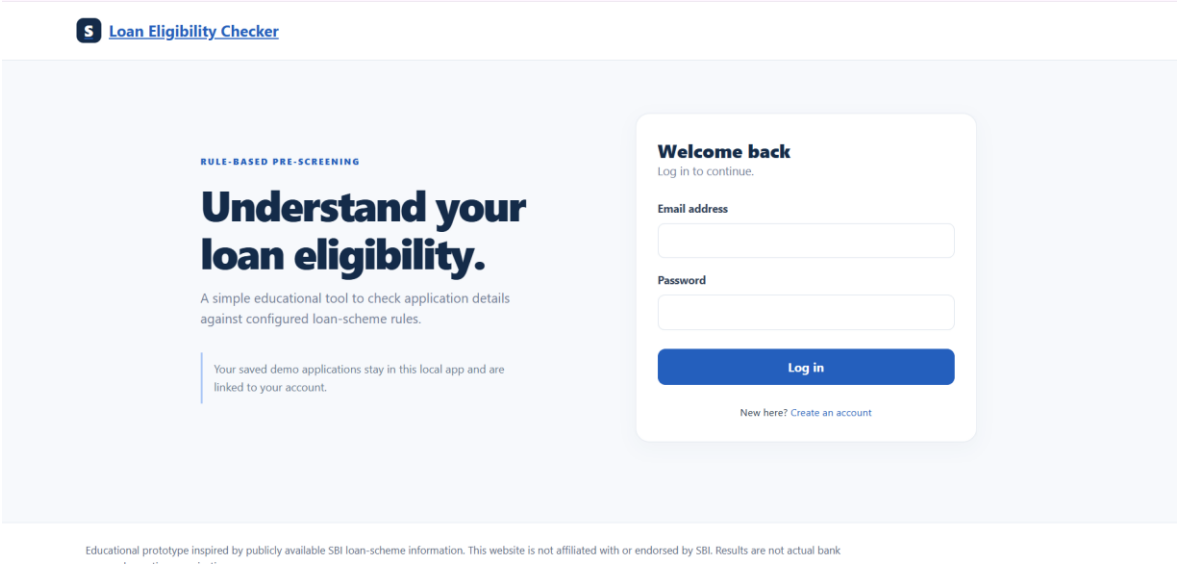


Fig. 5.1 Login page and registration page

5.2.2 Loan type selection page

Figure 5.2 displays the available loan categories, including Personal Loan, Home Loan, Car Loan and Education Loan. Users select the required category before entering their application details. The selected category determines which configured eligibility rules are applied.

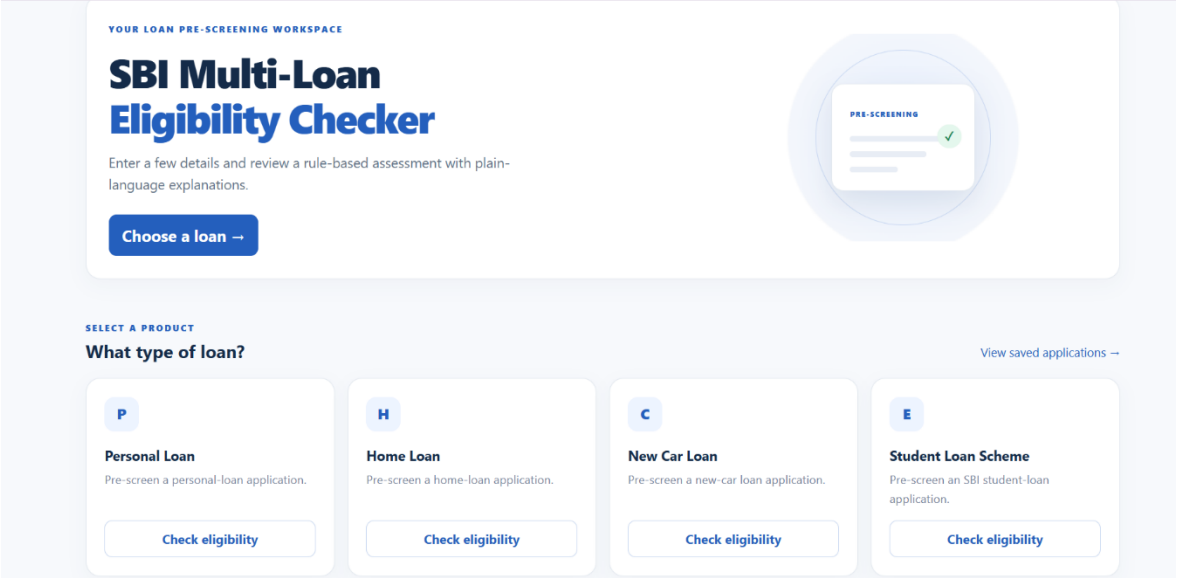


Fig. 5.2 Loan selection Interface

5.2.3 Loan Application Form

Figure 5.3 shows the form used to collect applicant and loan-related information. Details such as monthly income, employment status, credit score, requested loan amount and repayment period are entered as applicable. The validation module checks required fields and identifies missing or invalid information.

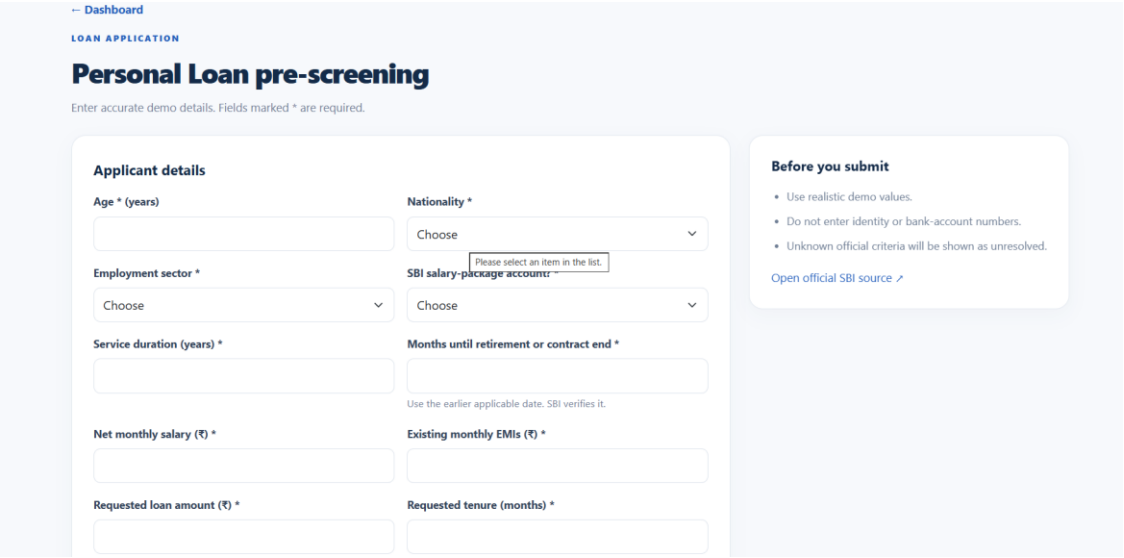


Fig. 5.3 Loan Application Form

5.2.4 AI Pre-Screening Result Page

Figure 5.4 displays the result generated by the rule-based inference engine. The result indicates Eligible, Ineligible or Incomplete according to the configured rules and submitted information. The explanation module provides reasons for the result and identifies conditions that require attention**.**

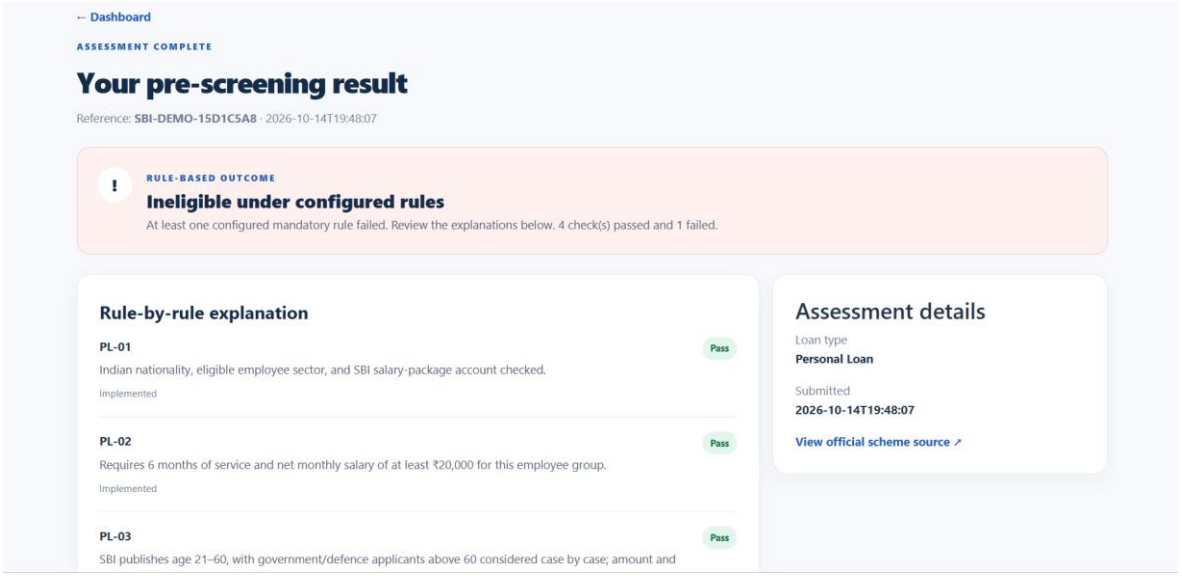


Fig. 5.4 AI-Based Loan Pre-Screening Result

5.2.5 Application History Page

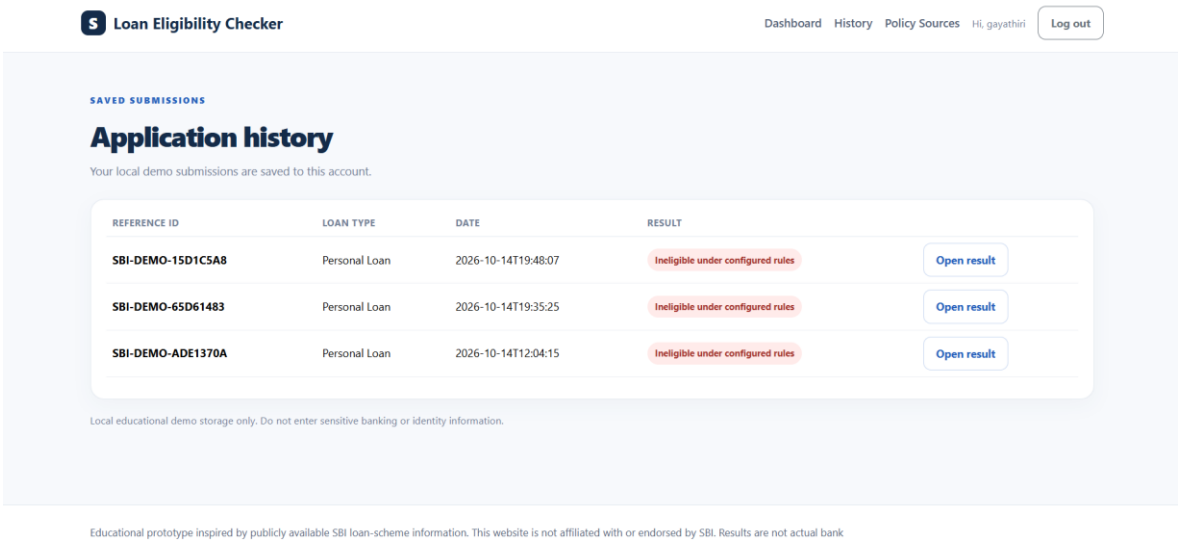


Fig. 5.5 Application History Interface

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Figure 5.5 shows the application history page. The system retrieves previously submitted applications and their saved screening results from the SQLite database. This feature allows users to review their earlier pre-screening assessments.

5.3 Rule-Based Evaluation Results

Table 5.3 summarises the expected behaviour of the system for different input conditions. Replace the illustrative cases with your actual test results after running the application.

Test Case	Input Condition	Expected Outcome	System Response
TC01	All applicable configured rules are satisfied	Eligible	Displays a positive pre-screening result and explanation
TC02	A defined disqualifying rule is triggered	Ineligible	Displays the failed condition and reason
TC03	Required applicant details are missing	Incomplete	Identifies missing information
TC04	An invalid input value is entered	Incomplete / Validation Error	Requests correction of the invalid information
TC05	A previous application is opened	Saved Result	Retrieves the stored application and result

These test cases assess the application's main functions, including validation, rule evaluation, explanation generation and database retrieval. The outcomes are educational pre-screening estimates and do not represent official bank approval or rejection.

5.4 Discussion

The implementation combines a web interface, Flask backend, rule-based inference engine and SQLite database into one application. The system evaluates different loan categories using configured IF–THEN rules and provides explanations for its results. Input validation helps identify missing or invalid information, while application history supports the retrieval of previous assessments.

The accuracy of the pre-screening results depends on the correctness and completeness of the configured rules. Therefore, the results should be interpreted as estimates produced by the educational prototype rather than actual lending decisions.

CHAPTER 6

CONCLUSION AND FUTURE WORK

6.1 Conclusion

This project developed an AI-Based Bank Loan Eligibility Checker using a Rule-Based Expert System. The system evaluates applicant details for Personal Loan, Home Loan, Car Loan and Education Loan using predefined IF–THEN rules. It validates the submitted information and generates Eligible, Ineligible or Incomplete pre-screening results based on the configured eligibility conditions.

The system provides explanations for screening results and stores user details, loan applications and results in an SQLite database. By integrating HTML, CSS, JavaScript, Bootstrap 5, Python Flask and SQLite, the project demonstrates how artificial intelligence techniques can support a consistent and explainable initial loan screening process. The system is an educational prototype and does not represent actual bank approval or rejection.

6.2 Future Work

- Integrate verified and updated eligibility criteria from official bank sources.
- Add support for more loan categories and their specific requirements.
- Improve the inference engine to evaluate complex financial conditions.
- Provide downloadable pre-screening reports in PDF format.
- Implement email notifications for screening results.
- Add charts and summaries to the application history dashboard.
- Test the system with more test cases to improve rule accuracy and system reliability.

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CHAPTER 7

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